The coefficients b_m are given by the formula

$$b_m = \frac{2j^2_{m,\nu}\int_0^1 \{tf'(t)-\nu f(t)\}J_{\nu+1}(j_{m,\nu}t)\,dt}{\{j^2_{m,\nu}-(\nu+1)^2\}J^2_{\nu+1}(j_{m,\nu})+j^2_{m,\nu}J'^2_{\nu+1}(j_{m,\nu})}$$

$$= \frac{2}{J^2_{\nu+1}(j_{m,\nu})}\int_0^1 \frac{d}{dt}\{t^{-\nu}f(t)\}\,t^{\nu+1}J_{\nu+1}(j_{m,\nu}t)\,dt$$

$$= \frac{2}{J^2_{\nu+1}(j_{m,\nu})}\left\{\left[tf(t)J_{\nu+1}(j_{m,\nu}t)\right]_0^1 - j_{m,\nu}\int_0^1 tf(t)J_\nu(j_{m,\nu}t)\,dt\right\}$$

$$= -j_{m,\nu}a_m,$$

provided that $$\left[tf(t)J_{\nu+1}(j_{m,\nu}t)\right]_0^1 = 0.$$

Sufficient conditions that this may be the case are

(i) $t^{\nu+2}f(t)\to 0$ as $t\to 0$,

(ii) $f(1-0)=0$,

(iii) $f(t)$ is continuous in the open interval in which $0<t<1$.

These conditions combined with the existence and absolute convergence of

$$\int_0^1 t^{\nu+\frac{1}{2}}\frac{d}{dt}\{t^{-\nu}f(t)\}\,dt$$

are sufficient to ensure the truth of (2) in any interval in which

$$f'(x)-(\nu/x)f(x)$$

has limited total fluctuation.

18·5. *The summability of Fourier-Bessel series.*

A consideration of the values of the coefficients in the Fourier-Bessel series associated with $f(x)$, combined with the expression of $T_n(t,x)$ as a contour integral, suggests that it is no easy matter to discuss by direct methods the question of the summability, by Cesàro's means, of the Fourier-Bessel expansion.

It is, however, very easy to investigate the summability when the method of Riesz* is used to 'sum' the series, and then the summability $(C1)$ can be inferred with the help of quite elementary analysis.

The expression which will be taken as the 'sum' of the series by the method of Riesz is

$$\lim_{n\to\infty}\sum_{m=1}^{n}\left(1-\frac{j_m}{A_n}\right)a_mJ_\nu(j_mx):$$

and when this limit exists, the Fourier-Bessel series will be said to be *summable* (R).

It is evident that

$$(1)\qquad \sum_{m=1}^{n}\left(1-\frac{j_m}{A_n}\right)a_mJ_\nu(j_mx)=\int_0^1 tf(t)\,T_n(t,x\,|\,R)\,dt,$$

* Cf. Hardy, *Proc. London Math. Soc.* (2) VIII. (1910), p. 309.

where

$$T_n(t, x \mid R) = \sum_{m=1}^{n} \left(1 - \frac{j_m}{A_n}\right) \frac{2J_\nu(j_m x) J_\nu(j_m t)}{J^2_{\nu+1}(j_m)}, \tag{2}$$

and so it will be convenient to discuss the properties of $T_n(t, x \mid R)$ after the manner of § 18·22 before we make further progress with the main problem.

18·51. *Theorems concerning* $T_n(t, x \mid R)$.

When $T_n(t, x \mid R)$ is defined by equation (2) of § 18·5, it is a symmetric function of t and x, and so we shall proceed to establish the properties of the function on the hypothesis that $0 \leqslant t \leqslant x \leqslant 1$, and we can then write down the corresponding properties when $0 \leqslant x \leqslant t \leqslant 1$ by interchanging t and x in the results already obtained.

We first observe that $T_n(t, x \mid R)$ is the sum of the residues of

$$\pi w \left(1 - \frac{w}{A_n}\right) \{J_\nu(w)\, Y_\nu(xw) - J_\nu(xw)\, Y_\nu(w)\} \frac{J_\nu(tw)}{J_\nu(w)}$$

at $j_1, j_2, j_3, \ldots, j_n$.

For brevity we write

$$w \{J_\nu(w)\, Y_\nu(xw) - J_\nu(xw)\, Y_\nu(w)\} \equiv \Phi(w, x),$$

and then it is obvious that, when* $t < x$,

$$\begin{aligned} T_n(t, x \mid R) &= \frac{1}{2i} \left[\int_{A_n - \infty i}^{A_n + \infty i} - \int_{-\infty i}^{\infty i} \right] \left(1 - \frac{w}{A_n}\right) \Phi(w, x) \frac{J_\nu(tw)\, dw}{J_\nu(w)} \\ &= \frac{1}{2i} \int_{A_n - \infty i}^{A_n + \infty i} \left(1 - \frac{w}{A_n}\right) \Phi(w, x) \frac{J_\nu(tw)\, dw}{J_\nu(w)} \\ &\quad + \frac{1}{2iA_n} \int_{-\infty i}^{\infty i} w\, \Phi(w, x) \frac{J_\nu(tw)\, dw}{J_\nu(w)}, \end{aligned}$$

since $\Phi(w, x)\, J_\nu(tw)/J_\nu(w)$ is an odd function of w.

We shall now obtain some upper bounds for

$$\Phi(w, x)\, J_\nu(tw)/J_\nu(w)$$

both when w is on the line joining $A_n - \infty i$ to $A_n + \infty i$, and when w is on the imaginary axis; the formulae which will be discussed are valid when $0 \leqslant x \leqslant 1$ and $0 \leqslant t \leqslant 1$, the sign of $x - t$ being immaterial.

To obtain these inequalities, we shall use series of ascending powers of w when $|w|$ is not large, and inequalities derived from the formulae of Chapter VII when $|w|$ is not small.

* When $t \geqslant x$, the integrals taken along the lines joining $\pm iB$ to $A_n \pm iB$ do not tend to zero as $B \to \infty$. There is no need to make an indentation at the origin, because $\Phi(w, x)$ is analytic at the origin.

We first deal with the factor $J_\nu(tw)/J_\nu(w)$. We observe that*

$$\left|\frac{J_\nu(tw)}{J_\nu(w)}\right| < \frac{k_1}{\sqrt{t}} \exp\{-(1-t)\,|\,I(w)\,|\} \tag{1}$$

when w is on either contour; this follows from inequalities of the type § 18·21 (9) when $|w|$ is not small, and from the ascending series when $|w|$ is not large (i.e. less than j_m).

We next consider $\Phi(w, x)$, which is equal to

$$\tfrac{1}{2}iw\{H_\nu^{(1)}(w)\,H_\nu^{(2)}(xw) - H_\nu^{(1)}(xw)\,H_\nu^{(2)}(w)\};$$

it is convenient to make two investigations concerning this function, the former being valid when $-\frac{1}{2} \leqslant \nu \leqslant \frac{1}{2}$, the second when $\nu \geqslant \frac{1}{2}$.

(I) The first investigation is quite simple. It follows from § 3·6 and § 7·33 that

$$|H_\nu^{(1)}(xw)| < \frac{k_2\,|e^{ixw}|}{|xw|^{\frac{1}{2}}}, \quad |H_\nu^{(2)}(xw)| < \frac{k_2\,|e^{-ixw}|}{|xw|^{\frac{1}{2}}} \tag{2}$$

for *all* the values of w and x under consideration when $-\frac{1}{2} \leqslant \nu \leqslant \frac{1}{2}$. Hence

$$\Phi(w, x) < \frac{k_2^2}{\sqrt{x}} \exp\{(1-x)\,|\,I(w)\,|\}. \tag{3}$$

(II) When $\nu \geqslant \frac{1}{2}$ and $|w|$ is not large, it is easy to deduce from the ascending series for $J_\nu(w)$, $Y_\nu(w)$, $J_\nu(xw)$ and $Y_\nu(xw)$ that

$$|\Phi(w, x)| < k_3\,|w|\,x^{-\nu}. \tag{4}$$

If $|w|$ is not small, we use the inequalities (deduced from § 7·33)

$$|H_\nu^{(1)}(w)| < \frac{k_2\,|e^{iw}|}{|w|^{\frac{1}{2}}}, \quad |H_\nu^{(2)}(w)| < \frac{k_2\,|e^{-iw}|}{|w|^{\frac{1}{2}}}, \tag{5}$$

together with the inequalities

$$\left.\begin{aligned} |H_\nu^{(1)}(xw)| &< k_4\{|xw|^{-\frac{1}{2}} + |xw|^{-\nu}\}\,|e^{ixw}|, \\ |H_\nu^{(2)}(xw)| &< k_4\{|xw|^{-\frac{1}{2}} + |xw|^{-\nu}\}\,|e^{-ixw}|. \end{aligned}\right\} \tag{6}$$

It follows from § 3·6 and § 7·33 that the inequalities (6) are true whether $|xw|$ is large or not. Hence,

$$|\Phi(w, x)| < k_2 k_4\{x^{-\frac{1}{2}} + x^{-\nu}\,|w|^{\frac{1}{2}-\nu}\} \exp\{(1-x)\,|\,I(w)\,|\}, \tag{7}$$

when $\nu \geqslant \frac{1}{2}$ and $|w|$ is large, whatever be the magnitude† of $|xw|$.

If we now combine the results contained in formulae (3), (4) and (7) we deduce that, whether $-\frac{1}{2} \leqslant \nu \leqslant \frac{1}{2}$ or $\nu \geqslant \frac{1}{2}$,

$$|\Phi(w, x)| < k_5(x^{-\frac{1}{2}} + x^{-\nu}) \exp\{(1-x)\,|\,I(w)\,|\}, \tag{8}$$

* It is supposed that the numbers k_1, k_2, k_3, ... are positive and independent of w, x and t: their values may, however, depend on the value of ν.

† Provided of course that $0 \leqslant x \leqslant 1$.

when w is any point of either contour and $0 \leqslant x \leqslant 1$. Hence, by (1), it follows that

$$(9) \qquad \left| \Phi(w,x) \frac{J_\nu(tw)}{J_\nu(w)} \right| < k_6 t^{-\frac{1}{2}} (x^{-\frac{1}{2}} + x^{-\nu}) \exp\{-(x-t)|I(w)|\},$$

when $0 \leqslant x \leqslant 1$ and $0 \leqslant t \leqslant 1$.

We now return to the integral formula for $T_n(t,x|R)$. If we replace w by $A_n \pm iv$ and $\pm iv$ in the first and second contour integrals respectively, we deduce that, when $0 \leqslant t < x \leqslant 1$,

$$|T_n(t,x|R)| < \frac{2k_6}{A_n\sqrt{t}} (x^{-\frac{1}{2}} + x^{-\nu}) \int_0^\infty v e^{-(x-t)v} dv = \frac{2k_6 (x^{-\frac{1}{2}} + x^{-\nu})}{A_n (x-t)^2 \sqrt{t}}.$$

We have consequently proved the two inequalities

$$(10) \qquad |T_n(t,x|R)| < \frac{2k_6 (x^{-\frac{1}{2}} + x^{-\nu})}{A_n (x-t)^2 \sqrt{t}} \qquad (0 \leqslant t < x \leqslant 1),$$

$$(11) \qquad |T_n(t,x|R)| < \frac{2k_6 (t^{-\frac{1}{2}} + t^{-\nu})}{A_n (x-t)^2 \sqrt{x}} \qquad (0 \leqslant x < t \leqslant 1).$$

It is to be remembered that k_6 is independent of x and t, so that we may make $|x-t|$ tend to zero, if we desire to do so.

One other pair of inequalities is required in order to discuss the behaviour of $T_n(t,x|R)$ when x and t are nearly equal. To obtain them, we write

$$T_n(t,x|R) = \frac{1}{2i} \int \left(1 - \frac{w}{A_n}\right) \Phi(w,x) \frac{J_\nu(tw)\,dw}{J_\nu(w)},$$

when $0 \leqslant t \leqslant x \leqslant 1$; in this integral the contour is taken to be a rectangle with vertices $\pm iA_n$, $A_n \pm iA_n$.

It is easy to see that (9) is satisfied whether w be on the horizontal sides or on the vertical sides of this rectangle: and the factor $1 - (w/A_n)$ does not exceed $\sqrt{2}$ in absolute value at any point of the contour.

Consequently the modulus of the integrand does not exceed

$$k_6 t^{-\frac{1}{2}} (x^{-\frac{1}{2}} + x^{-\nu}) \sqrt{2};$$

and since the length of the contour is $6A_n$, we infer that, when $0 \leqslant t \leqslant x \leqslant 1$,

$$(12) \qquad |T_n(t,x|R)| < \frac{3A_n k_6 (1 + x^{\frac{1}{2}-\nu})}{\sqrt{(\frac{1}{2}tx)}},$$

and similarly, when $0 \leqslant x \leqslant t \leqslant 1$,

$$(13) \qquad |T_n(t,x|R)| < \frac{3A_n k_6 (1 + t^{\frac{1}{2}-\nu})}{\sqrt{(\frac{1}{2}tx)}}.$$

The last four inequalities are sufficient to enable us to discuss adequately the summability (R) of Fourier-Bessel series. The reader will observe that the consideration of small values of x has increased the length of the analysis to an appreciable but not to an undue extent.

18·52. *The analogue of Fejér's theorem.*

We can now prove that *the existence and the absolute convergence of*

$$\int_0^1 t^{\frac{1}{2}} f(t)\,dt$$

are sufficient to ensure that the Fourier-Bessel series associated with $f(t)$ *is summable* (R) *at all points* x *of the open interval* $(0, 1)$ *at which the two limits* $f(x \pm 0)$ *exist. And the sum* (R) *of the series is*

$$\tfrac{1}{2}\{f(x+0)+f(x-0)\}.$$

This theorem is obviously the analogue of Fejér's theorem* concerning Fourier series.

Since† a series which is *convergent* is summable (R), it follows from § 18·35 that, when $0 < x < 1$,

$$\lim_{n\to\infty} \int_0^x t^{\nu+1} T_n(t, x \mid R)\,dt = \lim_{n\to\infty} \int_x^1 t^{\nu+1} T_n(t, x \mid R)\,dt$$
$$= \tfrac{1}{2} x^{\nu}.$$

Hence it follows that, when the limits $f(x \pm 0)$ exist, then

$$\lim_{n\to\infty} \int_0^x t^{\nu+1} T_n(t, x \mid R)\, x^{-\nu} f(x-0)\,dt + \lim_{n\to\infty} \int_x^1 t^{\nu+1} T_n(t, x \mid R)\, x^{-\nu} f(x+0)\,dt$$
$$= \tfrac{1}{2}\{f(x+0) + f(x-0)\}.$$

We are now in a position to consider the sum $S_n(x \mid R)$, defined as

$$\sum_{m=1}^{n} \left(1 - \frac{j_m}{A_n}\right) a_m J_\nu(j_m x) - \int_0^x t^{\nu+1} T_n(t, x \mid R)\, x^{-\nu} f(x-0)\,dt$$
$$- \int_x^1 t^{\nu+1} T_n(t, x \mid R)\, x^{-\nu} f(x+0)\,dt,$$

and we shall prove that it can be made arbitrarily small by taking n sufficiently large.

The sum $S_n(x \mid R)$ is equal to

$$\int_0^x t^{\nu+1} \{t^{-\nu} f(t) - x^{-\nu} f(x-0)\}\, T_n(t, x \mid R)\,dt$$
$$+ \int_x^1 t^{\nu+1} \{t^{-\nu} f(t) - x^{-\nu} f(x+0)\}\, T_n(t, x \mid R)\,dt.$$

Now, on the hypothesis that the limits $f(x \pm 0)$ exist, if we choose an arbitrary positive number ϵ, there exists a positive number‡ δ such that

$$\left\{\begin{array}{ll} |\, t^{-\nu} f(t) - x^{-\nu} f(x+0)\,| < \epsilon, & (x \leqslant t \leqslant x+\delta), \\ |\, t^{-\nu} f(t) - x^{-\nu} f(x-0)\,| < \epsilon, & (x \geqslant t \geqslant x-\delta). \end{array}\right.$$

We now choose a positive function of n, say $\sigma(n)$, which is less than δ for sufficiently large values of n, and divide the interval $(0, 1)$ into six parts by the points $x \pm \delta$, $x \pm \sigma(n)$, x.

* Cf. *Modern Analysis*, § 9·4. † Cf. *Modern Analysis*, § 8·43.

‡ It is convenient to take δ less than x and $1 - x$.

In the intervals $(0, x-\delta)$, $(x-\delta,\ x-\sigma(n))$ and also in the intervals $(x+\sigma(n), x+\delta)$, $(x+\delta, 1)$ we use inequalities of the form given in § 18·51 (10) and (11); and in the intervals $(x-\sigma(n), x)$, $(x, x+\sigma(n))$ we use inequalities of the form given in § 18·51 (12) and (13).

It is thus found that $|S_n(x\,|\,R)|$ does not exceed

$$\frac{2k_6(x^{-\frac{1}{2}}+x^{-\nu})}{A_n\delta^2}\int_0^{x-\delta}\left|t^{\frac{1}{2}}f(t)-t^{\nu+\frac{1}{2}}x^{-\nu}f(x-0)\right|dt$$

$$+\frac{2k_6\epsilon(x^{-\frac{1}{2}}+x^{-\nu})}{A_n}\left[\int_{x-\delta}^{x-\sigma(n)}\frac{t^{\nu+\frac{1}{2}}\,dt}{(x-t)^2}+\frac{3A_n^2}{\sqrt{2}}\int_{x-\sigma(n)}^{x}t^{\nu+\frac{1}{2}}\,dt\right]$$

$$+\frac{4k_6\epsilon}{A_n\sqrt{x}}\left[\frac{3A_n^2}{\sqrt{2}}\int_x^{x+\sigma(n)}dt+\int_{x+\sigma(n)}^{x+\delta}\frac{dt}{(x-t)^2}\right]$$

$$+\frac{2k_6}{A_n\delta^2\sqrt{x}}\int_{x+\delta}^{1}(t^{\nu+\frac{1}{2}}+t)\left|t^{-\nu}f(t)-x^{-\nu}f(x+0)\right|dt.$$

For any given value of ϵ (and therefore of δ), the first and last terms in this expression can be made arbitrarily small by taking n sufficiently large, on account of the convergence of

$$\int_0^1\left|t^{\frac{1}{2}}f(t)\right|dt.$$

The remaining terms do not exceed

$$\frac{2k_6\epsilon(3x^{-\frac{1}{2}}+x^{-\nu})}{A_n}\left\{\frac{1}{\sigma(n)}+\frac{3A_n^2\sigma(n)}{\sqrt{2}}\right\},$$

and, if we take $\sigma(n)\equiv 1/A_n$, this is independent of n, and it can be made as small as we please by taking ϵ sufficiently small initially.

We can therefore make the intermediate terms in the expression for $S_n(x\,|\,R)$ as small as we please by taking ϵ sufficiently small, and when this has been done, the first and last terms can be made as small as we please by taking n sufficiently large.

That is to say, $S_n(x\,|\,R)$ can be made arbitrarily small by taking n sufficiently large, so that

$$\lim_{n\to\infty}S_n(x\,|\,R)=0.$$

Hence

$$\lim_{n\to\infty}\sum_{m=1}^{n}\left(1-\frac{j_m}{A_n}\right)a_m J_\nu(j_m x)=x^{-\nu}f(x-0)\lim_{n\to\infty}\int_0^x t^{\nu+1}T_n(t, x\,|\,R)\,dt$$

$$+x^{-\nu}f(x+0)\lim_{n\to\infty}\int_x^1 t^{\nu+1}T_n(t, x\,|\,R)\,dt,$$

since the limits on the right exist.

Since each of the limits on the right is equal to $\frac{1}{2}x^{\nu}$, it has now been proved that

$$\sum_{m=1} a_m J_\nu(j_m x)$$

is summable (R) with sum $\frac{1}{2}\{f(x+0)+f(x-0)\}$ provided that the limits $f(x\pm 0)$ exist; and this is the theorem to be established.

As a corollary, the reader should be able to prove without difficulty that, if $f(t)$ is *continuous* in (a, b), the summability (R) is *uniform* throughout the interval in which $a+\Delta \leqslant x \leqslant b-\Delta$, where Δ is any positive number. Cf. § 18·25.

18·53. *Uniformity of summability of the Fourier-Bessel series near the origin.*

We shall now examine the *uniformity* of the summability (R) of the Fourier-Bessel expansion throughout an interval of which the origin is an end-point. It will be supposed that the expansion is modified by being multiplied throughout by $\sqrt{x}$, and it will then be proved that, *if* $t^{-\nu}f(t)$ *is continuous in the interval* $(0, b)$, *then the modified expansion is uniformly summable throughout* $(0, b-\Delta)$, *where* Δ *is any positive number.*

Given ϵ, we can now choose δ (less than Δ) so that

$$|\{t^{-\nu}f(t)-x^{-\nu}f(x)\}|<\epsilon$$

whenever $x-\delta \leqslant t \leqslant x+\delta$ and $t \geqslant 0$, provided that x lies in $(0, b-\Delta)$.

Since continuity involves uniformity of continuity, this choice of δ may be taken to be independent of x.

We now write

$$S_n(x\,|\,R)=\int_0^1 t^{\nu+1}\{t^{-\nu}f(t)-x^{-\nu}f(x)\}\,T_n(t, x\,|\,R)\,dt$$

and then examine $|x^{\frac{1}{2}}S_n(x\,|\,R)|$ after the manner of § 18·52.

We express $x^{\frac{1}{2}}S_n(x\,|\,R)$ as the sum of six integrals (some of which are to be omitted when $x<\delta$), and we see that $x^{\frac{1}{2}}S_n(x\,|\,R)$ does not exceed

$$\frac{2k_6(x^{\nu+\frac{1}{2}}+x)}{A_n\delta^2}\int_0^{x-\delta}|\{t^{-\nu}f(t)-x^{-\nu}f(x)\}|\,dt$$

$$+\frac{2k_6\epsilon(x^{\nu+\frac{1}{2}}+x)}{A_n}\left[\int_{x-\delta}^{x-\sigma(n)}\frac{dt}{(t-x)^2}+\frac{3A_n^2}{\sqrt{2}}\int_{x-\sigma(n)}^{x}dt\right]$$

$$+\frac{4k_6\epsilon}{A_n}\left[\frac{3A_n^2}{\sqrt{2}}\int_x^{x+\sigma(n)}dt+\int_{x+\sigma(n)}^{x+\delta}\frac{dt}{(t-x)^2}\right]$$

$$+\frac{4k_6}{A_n\delta^2}\int_{x+\delta}^1|\{t^{-\nu}f(t)-x^{-\nu}f(x)\}|\,dt.$$

In this formula any of the limits of integration which are negative are supposed to be replaced by zero.

Now this upper bound for $|x^{\frac{1}{2}}S_n(x\,|\,R)|$ does not exceed

$$\frac{4k_6}{A_n\delta^2}\int_0^1 \{t^{-\nu}f(t)-x^{-\nu}f(x)\}\,|\,dt+\frac{8k_6\epsilon}{A_n}\left\{\frac{1}{\sigma(n)}+\frac{3A_n{}^2\sigma(n)}{\sqrt{2}}\right\},$$

and, since $x^{-\nu}f(x)$ is bounded (because it is continuous), this can be made arbitrarily small by a choice of n which is independent of x.

Consequently $x^{\frac{1}{2}}S_n(x\,|\,R)$ tends to zero uniformly as $n\to\infty$.

Now it has already been shewn (§ 18·22) that

$$x^{\frac{1}{2}}\int_0^1 t^{\nu+1}T_n(t,x)\,dt$$

is uniformly convergent in $(0, 1-\Delta)$, and so, since uniformity of convergence involves uniformity of summability,

$$x^{\frac{1}{2}-\nu}f(x)\int_0^1 t^{\nu+1}T_n(t,x\,|\,R)\,dt$$

tends uniformly to $x^{\frac{1}{2}}f(x)$ in $(0, b-\Delta)$.

Hence, since $x^{\frac{1}{2}}S_n(x\,|\,R)$ tends to zero uniformly,

$$x^{\frac{1}{2}}\int_0^1 tf(t)\,T_n(t,x\,|\,R)\,dt$$

tends uniformly to $x^{\frac{1}{2}-\nu}f(x)\int_0^1 t^{\nu+1}T_n(t,x\,|\,R)\,dt$, i.e. to $x^{\frac{1}{2}}f(x)$ in $(0, b-\Delta)$.

It has therefore been proved that

$$\sum_{m=1}^{\infty} a_m x^{\frac{1}{2}}J_\nu(j_m x)$$

is uniformly summable (R) in $(0, b-\Delta)$ with sum $x^{\frac{1}{2}}f(x)$, provided that

$$\int_0^1 t^{\frac{1}{2}}f(t)\,dt$$

exists and is absolutely convergent, and that $t^{-\nu}f(t)$ is continuous in $(0, b)$.

18·54. *Methods of 'summing' Fourier-Bessel series.*

We shall now investigate various methods of summing the Fourier-Bessel series*

$$\sum_{m=0}^{\infty} a_m x^{\frac{1}{2}}J_\nu(j_m x)$$

on the hypotheses (i) that the limits $f(x\pm 0)$ exist, (ii) that

$$\int_0^1 t^{\frac{1}{2}}f(t)\,dt$$

exists and is absolutely convergent, and (iii) that the series is summable (R).

It conduces to brevity to write $f_m(x)$ in place of $a_m x^{\frac{1}{2}}J_\nu(x)$, so that $f_m(x)$ tends uniformly to zero (§ 18·27) as $m\to\infty$ when x lies in $(0, 1)$.

* The factor $x^{\frac{1}{2}}$ is inserted merely in order that the discussion may cover the investigation of uniformity of summability near the origin.

Consider first the limit

$$\lim_{n\to\infty} \sum_{m=1}^{n} \left(1-\frac{j_m}{j_n}\right) f_m(x)$$

which gives the most natural method (of Riesz' type) for summing the series.

Since $(j_n/A_n)\to 1$, it is evident that

$$\lim_{n\to\infty} \sum_{m=1}^{n} \left(\frac{A_n - j_m}{j_n}\right) f_m(x)$$

exists and is equal to

$$\lim_{n\to\infty} \sum_{m=1}^{n} \left(1-\frac{j_m}{A_n}\right) f_m(x).$$

Again, since $f_n(x) = o(1)$, it is easy to see that

$$\sum_{m=1}^{n} f_m(x) = o(n),$$

so that

$$\lim_{n\to\infty} \left(\frac{A_n - j_n}{j_n}\right) \sum_{m=1}^{n} f_m(x) = 0,$$

and therefore

$$\lim_{n\to\infty} \sum_{m=1}^{n} \left(1-\frac{j_m}{j_n}\right) f_m(x) = \lim_{n\to\infty} \sum_{m=1}^{n} \left(1-\frac{j_m}{A_n}\right) f_m(x);$$

the limit on the right exists in consequence of the hypotheses made at the beginning of the section.

Again, since

$$\frac{j_m}{j_n} - \frac{m}{n} = O\left(\frac{1}{n}\right),$$

whether m be $o(n)$ or $O(n)$, it follows that

$$\lim_{n\to\infty} \sum_{m=1}^{n} \left(\frac{j_m}{j_n} - \frac{m}{n}\right) f_m(x) = 0,$$

and so

$$\lim_{n\to\infty} \sum_{m=1}^{n} \left(1-\frac{m}{n}\right) f_m(x) = \lim_{n\to\infty} \sum_{m=1}^{n} \left(1-\frac{j_m}{j_n}\right) f_m(x).$$

Consequently the hypotheses that the limits $f(x\pm 0)$ exist $(0<x<1)$ and that the integral

$$\int_0^1 t^{\frac{1}{2}} f(t)$$

exists and is absolutely convergent *are sufficient to ensure that*

$$\sum_{m=1}^{\infty} a_m x^{\frac{1}{2}} J_\nu(j_m x)$$

is summable $(C\,1)$ *with sum* $\frac{1}{2}x^{\frac{1}{2}}\{f(x+0)+f(x-0)\}$.

By the same reasoning, if $f(x)$ is continuous in (a, b), the summability $(C\,1)$ is *uniform* in $(a+\Delta, b-\Delta)$; and, if $a=0$ and $t^{-\nu}f(t)$ has a limit as $t\to 0$, the summability $(C\,1)$ is *uniform* in $(0, b-\Delta)$.

18·55. *Uniformity of convergence of the Fourier-Bessel expansion near the origin.*

We can now prove, by using Hardy's convergence theorem*, that, *if* $t^{\frac{1}{2}} f(t)$ *has limited total fluctuation in* $(0, b)$, while $f(t)$ is also subject to the conditions of § 18·53, then

$$\sum_{m=1}^{\infty} a_m x^{\frac{1}{2}} J_\nu(j_m x)$$

is *uniformly convergent* in $(0, b-\Delta)$ with sum $x^{\frac{1}{2}} f(x)$.

Let $h(t)$ be an auxiliary function defined to be equal to $f(t)$ in $(0, b)$ and equal to zero in $(b, 1)$; and let the Fourier-Bessel series associated with $h(t)$ be

$$\sum_{m=1}^{\infty} \alpha_m J_\nu(j_m x).$$

Then, by § 18·54, $\sum_{m=1}^{\infty} a_m x^{\frac{1}{2}} J_\nu(j_m x)$ is *uniformly summable* $(C1)$ throughout $(0, b-\Delta)$ with sum $x^{\frac{1}{2}} f(x)$, and, by Sheppard's theorem (§ 18·27), $\alpha_m / \sqrt{j_m}$ is $O(1/m)$, while $(j_m x)^{\frac{1}{2}} J_\nu(j_m x)$ is a bounded function of x and m. Hence, by Hardy's convergence theorem,

$$\sum_{m=1} \alpha_m x^{\frac{1}{2}} J_\nu(j_m x)$$

is *uniformly convergent* throughout $(0, b-\Delta)$, with sum $x^{\frac{1}{2}} f(x)$.

Again

$$\sum_{m=1}^{n} (a_m - \alpha_m) x^{\frac{1}{2}} J_\nu(j_m x) = x^{\frac{1}{2}} \int_b^1 t f(t)\, T_n(t, x)\, dt,$$

and this tends uniformly to zero in $(0, b-\Delta)$ as $n \to \infty$ by an analogue of the Riemann-Lebesgue lemma (§ 18·23).

Hence $\sum_{m=1}^{n} a_m x^{\frac{1}{2}} J_\nu(j_m x)$ tends uniformly to the sum $x^{\frac{1}{2}} f(x)$ in $(0, b-\Delta)$ as $n \to \infty$; and this is the theorem to be established.

18·56. *Summability of Dini series.*

Except when $x = 1$, the summability $(C1)$ of the Dini series associated with $f(t)$ may be inferred by combining the results of § 18·33 and §§ 18·51—18·53.

The summability $(C1)$ may, however, be established independently† for all points x such that $0 < x \leqslant 1$ by replacing A_n and the functions $J_\nu(w)$ and $Y_\nu(w)$, which occur in § 18·5, by D_n and the functions $wJ_\nu'(w) + HJ_\nu(w)$ and $wY_\nu'(w) + HY_\nu(w)$ respectively; the details of the analysis may be left to the reader, and he will find that when $x = 1$ the expression $\frac{1}{2}\{f(x+0) + f(x-0)$ must be replaced by $f(1-0)$.

* Cf. *Modern Analysis*, § 8·5.

† Of course on the hypotheses concerning $f(t)$ which were assumed in § 18·53.

The uniformity of the summability in the interval $(a+\Delta, 1)$ when $f(x)$ is continuous in $(a, 1)$ may be dealt with in the same way as the uniformity of convergence was dealt with in §§ 18·33, 18·35.

The summability of Dini series (and of Fourier-Bessel series) by a modification of Abel's method is of some physical importance. Thus, in Fourier's* problem of the Conduction of Heat in an infinite solid cylinder of radius unity, the temperature v at distance r from the axis satisfies the equation

$$\frac{dv}{dt} = k\left\{\frac{d^2v}{dr^2} + \frac{1}{r}\frac{dv}{dr}\right\},$$

with the boundary condition

$$\left[\frac{dv}{dr} + Hv\right]_{r=1} = 0,$$

if the initial distribution of heat is symmetrical.

Normal solutions of the differential equation satisfying the boundary condition are

$$J_0(\lambda_m r)\exp(-k\lambda_m^2 t),$$

and so the temperature v is given by the series†

$$\sum_{m=1}^{\infty} b_m J_0(\lambda_m r)\exp(-k\lambda_m^2 t),$$

where the coefficients b_m are to be determined from the consideration that

$$\sum_{m=1}^{\infty} b_m J_0(\lambda_m r)$$

is the Dini series associated with the initial temperature $f(r)$. It is evident that the initial temperature is expressible as

$$\lim_{t\to+0}\sum_{m=1}^{\infty} b_m J_0(\lambda_m r)\exp(-k\lambda_m^2 t);$$

and this limit exists when the Dini series is summable (R).

18·6. *The uniqueness of Fourier-Bessel series and Dini series.*

It has been shewn by Young‡ that the existence and the absolute convergence of

$$\int_0^1 t^{\frac{1}{2}} f(t)\,dt$$

are sufficient to ensure that *if all the coefficients a_m of the Dini series (or the Fourier-Bessel series) associated with $f(t)$ are zero, then the function $f(t)$ must be a null-function.*

* *La Théorie Analytique de la Chaleur* (Paris, 1822), §§ 306–320. Cf. Rayleigh, *Phil. Mag.* (6) XII. (1906), pp. 106—107 [*Scientific Papers*, V. (1912), pp. 338—339]; and Kirchhoff, *Berliner Sitzungsberichte*, 1883, pp. 519—524.

† In this physical problem, $H > 0$, and so there is no initial term to be inserted.

‡ *Proc. London Math. Soc.* (2) XVIII. (1920), pp. 174—175.